

MATH 1564 Notes: Transposes, Inverses, and Elementary Matrices

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1 Transpose and Symmetric Matrices

Definition (Transpose). *If A is an $m \times n$ matrix, then A^T is the matrix whose i th column is the i th row of A .*

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}.$$

Theorem (Transpose rules). *For matrices of compatible sizes and a scalar r ,*

1. $(A^T)^T = A$;
2. $(A + B)^T = A^T + B^T$;
3. $(rA)^T = rA^T$;
4. $(AB)^T = B^T A^T$.

Definition (Symmetric matrix). *A matrix A is symmetric if $A^T = A$.*

An $n \times n$ symmetric matrix has

$$\frac{n^2 - n}{2} + n = \frac{n^2 + n}{2} = \frac{n(n + 1)}{2}$$

independently chosen entries: the entries on and above the main diagonal determine the remaining entries.

Proof that $(AB)^T = B^T A^T$. The ji entry of $(AB)^T$ is the ij entry of AB :

$$(AB)_{ij} = a_{i1}b_{1j} + \cdots + a_{in}b_{nj}.$$

Recall that the ji entry of CD is the product of row j of C with column i of D . Thus the ji entry of $B^T A^T$ is

$$\begin{bmatrix} b_{1j} & \cdots & b_{nj} \end{bmatrix} \begin{bmatrix} a_{i1} \\ \vdots \\ a_{in} \end{bmatrix} = b_{1j}a_{i1} + \cdots + b_{nj}a_{in},$$

which is the same scalar. Therefore $(AB)^T = B^T A^T$. □

For example,

$$(ABCD)^T = D^T C^T B^T A^T.$$

2 Invertible Matrices

Definition (Invertible / nonsingular). *An $n \times n$ matrix A is invertible (or nonsingular) if there is an $n \times n$ matrix C such that*

$$CA = I_n = AC.$$

We denote C by A^{-1} .

Theorem (Uniqueness of an inverse). *The inverse A^{-1} is uniquely determined by A .*

Proof. Suppose C and B are both inverses of A . Then

$$C = C(AB) = (CA)B = IB = B.$$

□

2.1 The 2×2 inverse formula

The matrix

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

is invertible exactly when $ad - bc \neq 0$, and then

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Indeed,

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} = \begin{bmatrix} ad - bc & ab - ba \\ cd - dc & da - cb \end{bmatrix} = (ad - bc)I_2,$$

and multiplication in the other order gives the same result. For example,

$$\begin{bmatrix} 2 & 5 \\ -3 & -7 \end{bmatrix}^{-1} = \frac{1}{-14 + 15} \begin{bmatrix} -7 & -5 \\ 3 & 2 \end{bmatrix} = \begin{bmatrix} -7 & -5 \\ 3 & 2 \end{bmatrix}.$$

Theorem (Rules for inverses). *If A and B are invertible $n \times n$ matrices, then*

$$\begin{aligned} (A^{-1})^{-1} &= A, \\ (AB)^{-1} &= B^{-1}A^{-1}, \\ (A^T)^{-1} &= (A^{-1})^T. \end{aligned}$$

Proof. For the first rule, $A^{-1}A = I = AA^{-1}$. For the second,

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AIA^{-1} = I,$$

and similarly $B^{-1}A^{-1}AB = I$. For the third,

$$A^T(A^{-1})^T = (A^{-1}A)^T = I^T = I,$$

and similarly $(A^{-1})^T A^T = I$.

□

Useful consequences include

$$\begin{aligned}(ABCD)^{-1} &= D^{-1}C^{-1}B^{-1}A^{-1}, \\ (PAP^{-1})^{-1} &= (P^{-1})^{-1}A^{-1}P^{-1} = PA^{-1}P^{-1}, \\ (A^3)^{-1} &= (AAA)^{-1} = A^{-1}A^{-1}A^{-1} = (A^{-1})^3, \\ A^0 &= I.\end{aligned}$$

3 Computing an Inverse and Solving Systems

Theorem. *An $n \times n$ matrix A is invertible if and only if $\text{RREF}(A) = I_n$. If A is invertible, row reduction of the augmented matrix produces*

$$[A \mid I_n] \longrightarrow [I_n \mid A^{-1}].$$

Example. To compute the inverse of

$$A = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ 0 & 0 & 1 \end{bmatrix},$$

row-reduce:

$$\left[\begin{array}{ccc|ccc} 0 & 1 & 2 & 1 & 0 & 0 \\ 1 & 0 & 3 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right] \longrightarrow \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 0 & 1 & -3 \\ 0 & 1 & 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array} \right].$$

Hence

$$A^{-1} = \begin{bmatrix} 0 & 1 & -3 \\ 1 & 0 & -2 \\ 0 & 0 & 1 \end{bmatrix}.$$

If A^{-1} is known, solve $Ax = b$ by multiplying on the left by A^{-1} :

$$A^{-1}Ax = A^{-1}b \implies Ix = A^{-1}b \implies x = A^{-1}b.$$

In practice, it is usually faster to solve $Ax = b$ directly by row reduction: it uses fewer operations and a smaller augmented matrix. Computing A^{-1} amounts to solving the n systems

$$Ax = e_i \quad (i = 1, \dots, n).$$

Theorem (Invertibility and systems). *Let A be square. The following are equivalent:*

1. A is invertible;
2. $Ax = b$ is consistent for every b ;
3. $Ax = 0$ has only the zero solution.

Notes on the equivalences. Statement 2 holds exactly when A has a pivot in every row. Statement 3 holds exactly when A has a pivot in every column. Since A is square, these are equivalent.

For $1 \Rightarrow 3$, if $Ax = 0$, then

$$A^{-1}Ax = A^{-1}0 = 0,$$

so $x = 0$. Also, an invertible $n \times n$ matrix has n pivots. □

4 Why the Inverse Algorithm Works

Suppose an invertible $n \times n$ matrix A is transformed to I_n by row operations. Each row operation is left multiplication by an elementary matrix, so for elementary matrices $E_1, \dots, E_k$,

$$E_k \cdots E_1 A = I_n.$$

Each E_i is invertible. Therefore

$$A = E_1^{-1} E_2^{-1} \cdots E_k^{-1},$$

which shows that A is a product of invertible matrices and hence is invertible. In addition,

$$A^{-1} = (E_1^{-1} E_2^{-1} \cdots E_k^{-1})^{-1} = E_k \cdots E_1.$$

If A has n pivots, then row reduction of $[A \mid b]$ produces $[I_n \mid d]$. Thus

$$Ax = b \iff I_n x = d.$$

Example.

$$\begin{bmatrix} 1 & 2 \\ 3 & 2 \end{bmatrix} x = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Theorem. *If A is invertible, then row reduction of $[A \mid I_n]$ gives $[I_n \mid A^{-1}]$.*

Proof. As above, row reduction of

$$[A \mid e_1 \cdots e_n]$$

gives

$$[I_n \mid d_1 \cdots d_n].$$

For each i , the system $Ax = e_i$ reduces to $I_n x = d_i$. Therefore

$$d_i = x = A^{-1} e_i,$$

so d_i is the i th column of A^{-1} . □

5 Elementary Matrices

Definition. An elementary matrix is a matrix obtained from I by a single row operation.

For 2×2 matrices, examples are

$$\begin{array}{ll} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} & \text{swap rows,} \\ \begin{bmatrix} 1 & 0 \\ 0 & c \end{bmatrix} & \text{multiply row 2 by } c, \\ \begin{bmatrix} c & 0 \\ 0 & 1 \end{bmatrix} & \text{multiply row 1 by } c, \\ \begin{bmatrix} 1 & 0 \\ c & 1 \end{bmatrix} & \text{replace row 2 by row 2 plus } c \text{ times row 1.} \end{array}$$

Elementary matrices encode row operations by left multiplication. For example,

$$\begin{aligned} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} &= \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix}, \\ \begin{bmatrix} 1 & 0 \\ 0 & c \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} &= \begin{bmatrix} 1 & 2 \\ 3c & 4c \end{bmatrix}, \\ \begin{bmatrix} c & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} &= \begin{bmatrix} c & 2c \\ 3 & 4 \end{bmatrix}, \\ \begin{bmatrix} 1 & 0 \\ c & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} &= \begin{bmatrix} 1 & 2 \\ c+3 & 2c+4 \end{bmatrix}. \end{aligned}$$

Elementary matrices are invertible because every elementary row operation can be reversed. If $E_1 A$ is the result of one row operation and E_2 encodes its reverse, then

$$E_2 E_1 A = A, \quad \text{so} \quad E_2 E_1 = I.$$

Example. To identify the elementary matrix for

$$\begin{bmatrix} 1 & 1 & 1 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \longrightarrow \begin{bmatrix} 1 & 1 & 1 \\ 0 & 3 & 2 \\ 0 & 0 & 1 \end{bmatrix},$$

observe that the operation is $R_2 \leftarrow R_2 + 2R_1$. Hence

$$E = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix},$$

and indeed

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \\ -2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 3 & 2 \\ 0 & 0 & 1 \end{bmatrix}.$$

6 One-Sided Inverses

Left inverse. C is a *left inverse* of A if $CA = I$.

Right inverse. C is a *right inverse* of A if $AC = I$.

For example, a left inverse of

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

is

$$\begin{bmatrix} 1 & 0 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1 \end{bmatrix} = I_1.$$